BRAVO Lab

Brazilian Risk, Allocation, Volatility & Options Lab

Executive Report

v0.1.2

Brazilian Equity Risk, Volatility Transmission, Synthetic Protection Logic, BSTI Policy Selection, and Portfolio-Governance Diagnostics.

CREATED BY

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Research, portfolio risk, derivatives overlays, and market transmission analysis

Research prototype - Not investment advice

NAVIGATION

Contents

The report is organized as a decision review: current signal, evidence stack, visual diagnostics, validation layer, policy rule, implementation limits, and final decision read.

EXECUTIVE SUMMARY

What BRAVO Lab is deciding

BRAVO Lab is a portfolio-governance research prototype for Brazilian market risk. Its purpose is not to predict the market. Its purpose is to convert data, stress signals, synthetic option overlays, and active-risk diagnostics into a structured decision conversation.

The practical question is simple and valuable: when Brazilian risk changes state, should the portfolio remain passive, harvest volatility income through covered calls, protect downside through collars, or wait because the signal is not persistent enough? BRAVO gives that question a reproducible evidence stack instead of leaving it to intuition.

The project demonstrates the full chain from market data to decision language: data ingestion, volatility and drawdown diagnostics, regime classification, synthetic option-overlay comparison, active-risk measurement, multi-asset stress transmission, BSTI construction, threshold validation, calibration, policy selection, and implementation limits. That is the value of the report: it shows a working research pipeline, not a decorative backtest.

The current read is deliberately conservative. The latest BSTI state is calm, the dominant pressure channel is Brazil drawdown pressure, and the current policy action is passive Brazil equity. The important insight is that the system is not forcing action. It is creating discipline: monitor the signal, check persistence, compare policy behavior, and only escalate to income or protection overlays when evidence justifies it.

Decision field	Current project read
BSTI score	13.47
BSTI regime	Calm
Dominant pressure channel	Brazil Drawdown Pressure
Current policy action	Passive Brazil Equity
Dominant historical policy choice	Passive Brazil Equity (61.59%)
BSTI policy annualized active return	5.45%
Tracking error	10.18%
Information ratio	0.54
Max drawdown	-25.01%
Data window	2014-01-02 to 2026-06-09

DECISION LANGUAGE

Glossary for the report

The report uses portfolio and risk language because BRAVO is designed to support decisions, not simply display charts. These terms make the logic readable before the reviewer enters the technical repository.

BSTI. Brazil Stress Transmission Index. A 0 to 100 composite stress index built from Brazil-related pressure channels.

BSTI regime. The interpretation of the BSTI score: calm, fragile, stress, or extreme stress.

Dominant pressure channel. The risk source contributing most strongly to the current stress state.

Stress breadth. How many pressure channels are active at the same time.

Covered call. An income overlay that sells upside participation in exchange for option premium.

Collar. A protection overlay that combines downside protection with limited upside participation.

Tracking error. How far an overlay deviates from passive Brazilian equity exposure.

Information ratio. How much active return is generated per unit of tracking error.

Governance score. A calibration score used to compare whether thresholds and channel weights produce useful decision signals.

Synthetic overlay. A research approximation using option-pricing assumptions. It is not yet live B3 option-chain evidence.

FRONT-OFFICE DECISION MEMO

Current decision read

This is the decision entrance of BRAVO. It shows what the model reads now, which policy action is implied, which evidence supports the signal, and what a committee must check before any action is considered.

BRAVO Lab Front-Office Executive Memo

Purpose: portfolio-governance decision support for Brazilian risk, volatility transmission, and overlay policy selection.

Decision read

BRAVO Lab currently reads Brazilian risk through a BSTI score of 13.47, classified as Calm, with Brazil Drawdown Pressure as the dominant pressure channel. The current BSTI policy action is Passive Brazil Equity.

The model is not demanding an option overlay. The committee should still monitor whether stress is beginning to migrate into warning state.

Portfolio action snapshot

Item	Current read
Current BSTI score	13.47
Current BSTI regime	Calm
Dominant pressure channel	Brazil Drawdown Pressure
Current policy action	Passive Brazil Equity
Dominant historical policy choice	Passive Brazil Equity (61.59%)
BSTI policy annualized active return	5.45%
BSTI policy tracking error	10.18%
BSTI policy information ratio	0.54
BSTI policy max drawdown	-25.01%

Evidence stack

Question	Evidence read
Which strategy had the best information ratio?	Bsti Policy Overlay
Which strategy had the best drawdown profile?	Collar
Which strategy had the highest annualized return?	Bsti Policy Overlay
How persistent are warning states?	Average warning duration: 1.88 observations
How persistent are stress states?	Average stress duration: 1.71 observations
How often do warnings escalate?	Warning-to-stress escalation rate: 24.17%
Which BSTI calibration is strongest?	Balanced, 63d horizon, threshold 10.00, governance score 0.50

Risk committee agenda

- 1. Confirm whether the current BSTI state is persistent or only a temporary pressure print.
- 2. Check whether the implied policy action matches the committee's drawdown tolerance.
- 3. Compare the BSTI policy overlay against passive equity, covered calls, collars, and the local stress-aware overlay.

- 4. Review whether the selected action is justified after transaction costs, liquidity, taxes, and implementation constraints.
- 5. Decide whether the stress state requires monitoring, hedge discussion, or actual overlay activation.

Implementation warning

This memo is a decision-support layer, not an investment recommendation. The evidence is generated from public market data, synthetic option-premium logic, transparent stress classification, and reproducible CSV outputs. Real B3 option-chain data, liquidity filters, tax effects, and portfolio mandate constraints must be added before production use.

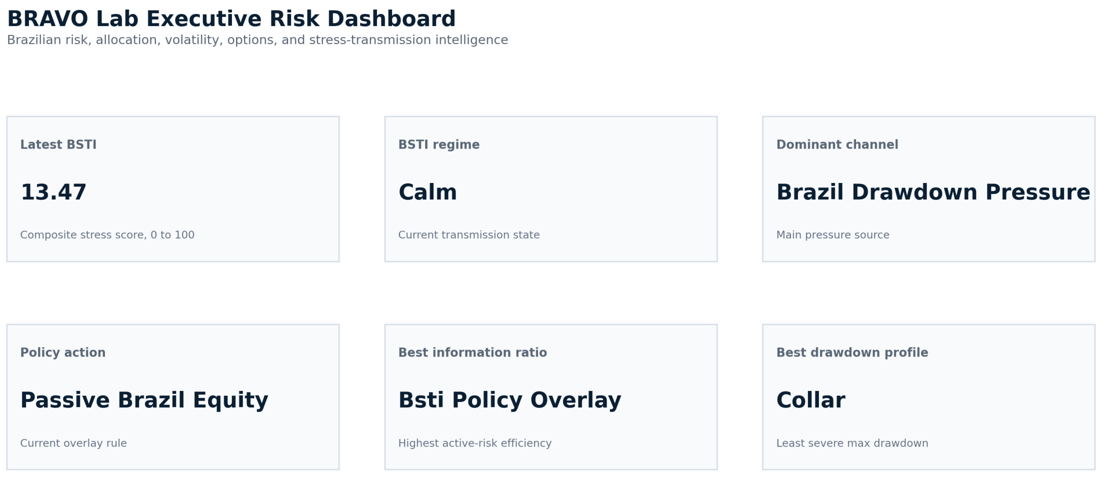
VISUAL EVIDENCE

Visual evidence and decision interpretation

The visual layer is the fastest inspection layer of BRAVO. It shows whether the project is merely calculating numbers or actually translating Brazilian market stress into portfolio decisions. The sequence matters: first the current decision read, then the strategy path, then drawdown behavior, then the stress index, then the policy rule, then persistence and calibration.

The value is not the image itself. The value is the decision story each image supports: whether the system preserves beta, when it harvests income, when it protects downside, whether stress persists, and whether the BSTI rule is auditable enough to guide a real risk discussion.

Executive risk dashboard

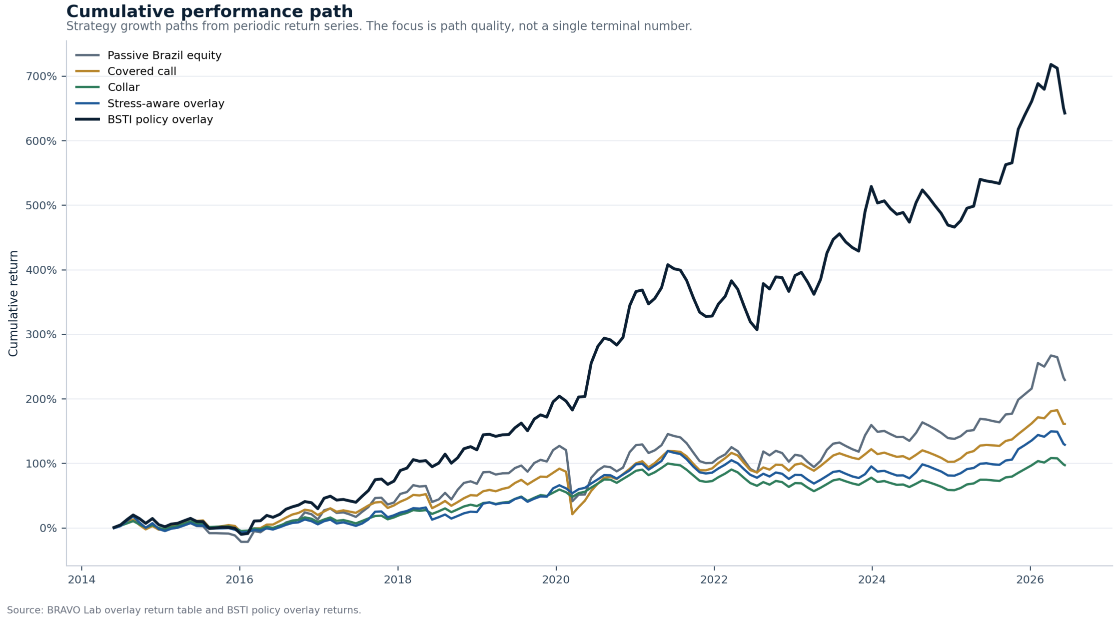


Source: BRAVO Lab processed outputs. Dashboard is generated automatically from reproducible CSV evidence files.

The dashboard is the decision surface of the project. It shows the current BSTI score, regime, dominant pressure channel, policy action, strongest information-ratio strategy, and strongest drawdown profile in one view.

This image matters because it converts many processed outputs into a portfolio question: does the current state justify passive exposure, income capture, protection, or only monitoring? In the current run, the signal is calm and the policy action is passive, which means the model is disciplined enough not to force an overlay when the evidence does not demand it.

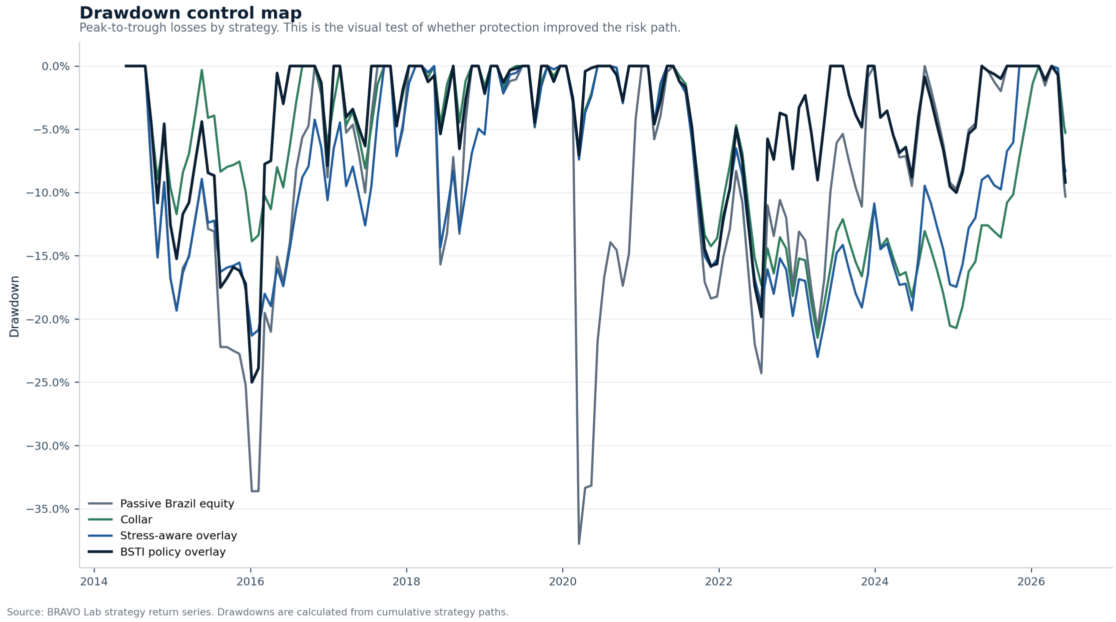
Cumulative performance path



This chart compares passive Brazilian equity, covered calls, collars, stress-aware switching, and the BSTI policy overlay under the same market history.

The professional question is not which line looks most impressive. It is whether the improvement in path behavior is worth the active risk, transaction costs, liquidity limits, tax effects, and synthetic option assumptions. This is where BRAVO moves from return display to portfolio-governance reasoning.

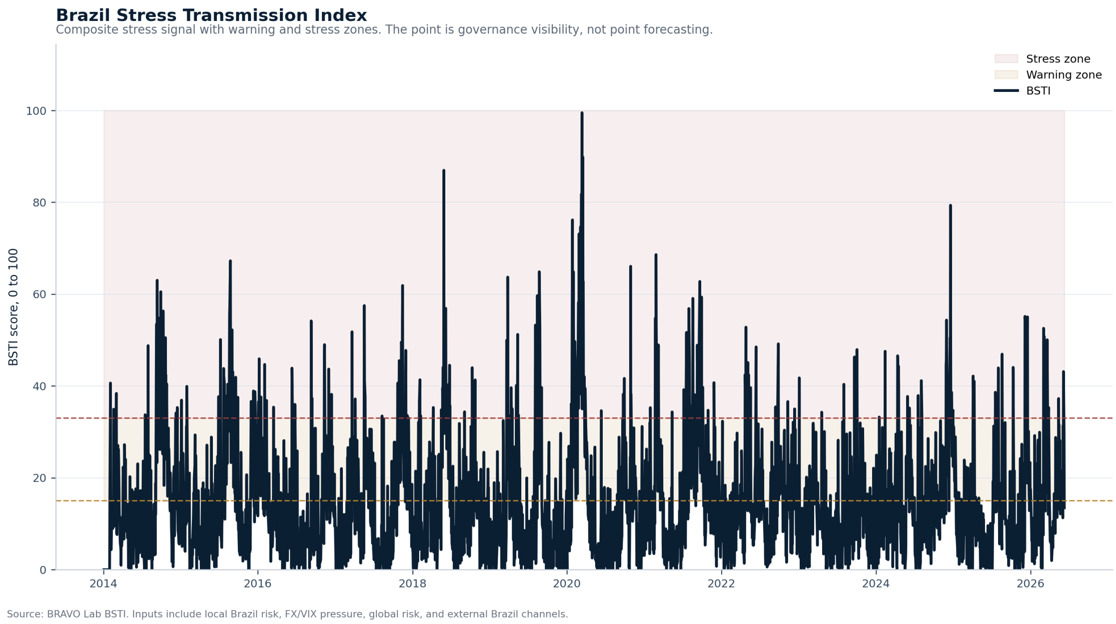
Drawdown control map



This chart shows how much each strategy loses from its previous peak. That is the practical test of protection because drawdowns are where portfolio discipline becomes visible.

The collar is judged by whether it reduces left-tail pain. The covered call is judged by whether the income earned is worth the risk of capped recovery. This chart is important because it shows the cost of ignoring downside behavior when comparing overlays only by return.

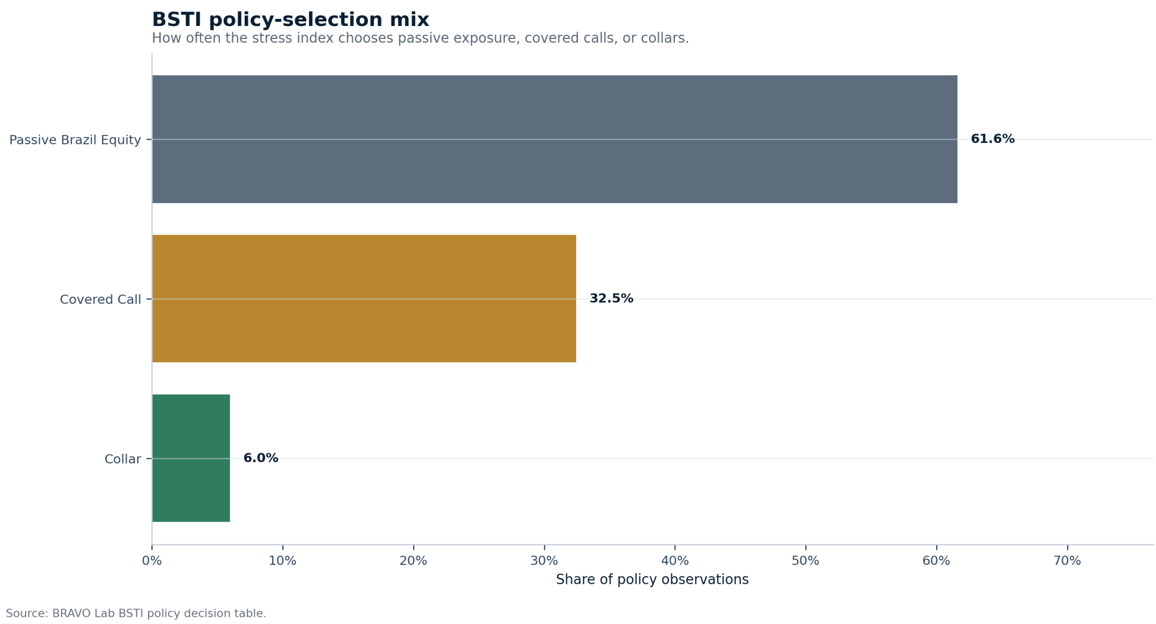
Brazil Stress Transmission Index



BSTI translates several Brazil-related stress channels into one monitorable 0 to 100 signal.

The chart is not a forecast. It is a governance signal. When the index moves toward warning or stress, the committee has a reason to inspect persistence, dominant channel, and overlay choice. That is the value: it turns scattered risk inputs into a disciplined escalation process.

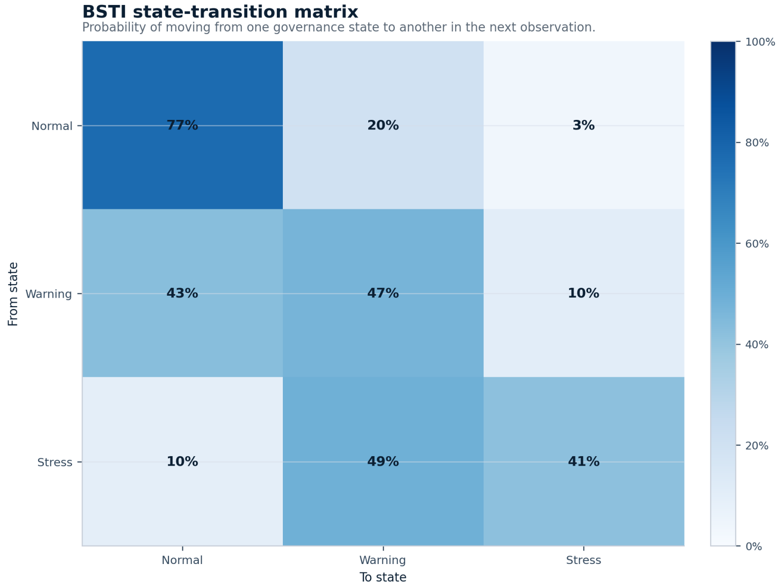
BSTI policy-selection mix



This chart shows how often the BSTI rule selects passive equity, covered calls, or collars.

A disciplined policy should not trade all the time. The mix shows whether BRAVO mostly preserves beta, selectively harvests volatility income, or moves toward protection during stress.

BSTI state-transition matrix

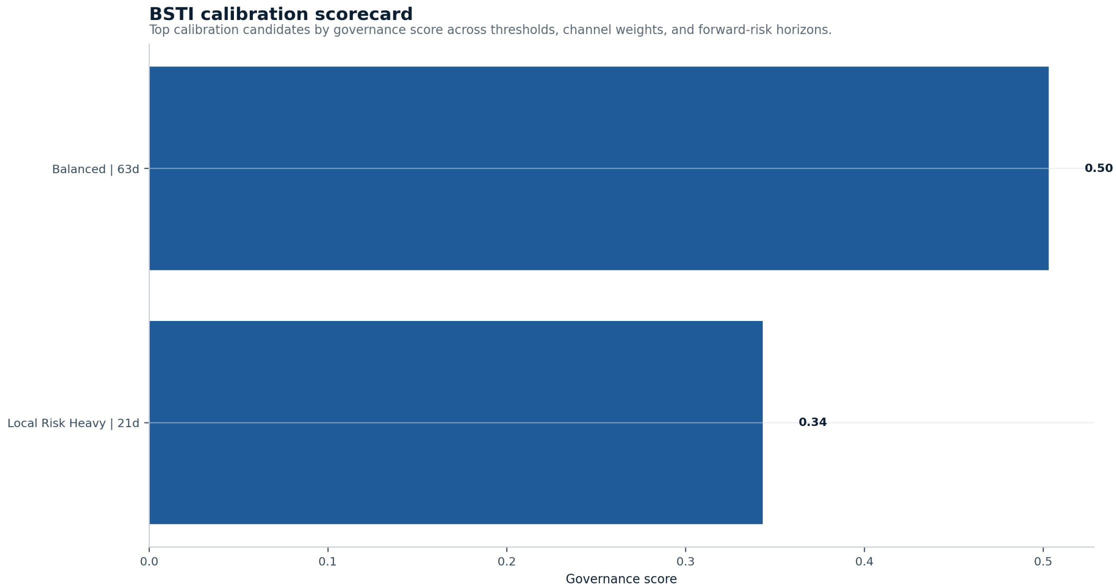


Source: BRAVO Lab BSTI transition diagnostics.

This chart shows whether normal, warning, and stress states persist or quickly reverse.

Persistence matters because a one-period warning should not trigger the same response as a durable stress state. This is the evidence behind escalation discipline.

BSTI calibration scorecard



Source: BRAVO Lab BSTI calibration grid and best-by-horizon calibration summary.

This chart compares threshold and weighting choices for the BSTI signal.

The value is auditability. The stress index is not treated as magic; it is tested against governance criteria such as precision, recall, signal frequency, and drawdown relevance.

EVIDENCE STACK

Original BRAVO evidence, styled and explained

The next sections preserve the original report's substance while improving readability. The evidence remains the same project logic: market state, data provenance, regime diagnosis, baseline risk, synthetic overlays, active risk, multi-asset stress, BSTI validation, calibration, policy selection, persistence, implementation drag, option attribution, and final limits.

3. Portfolio Question

BRAVO begins with a portfolio decision, not a chart. The question is whether Brazilian equity exposure should remain passive, monetize volatility through covered calls, protect downside through collars, or wait for stronger evidence.

The core question is not whether passive equity, covered calls, collars, or stress-aware switching are better in isolation. The correct question is regime-dependent.

When Brazilian equity risk changes state, should the portfolio keep full beta, sell volatility through covered calls, pay for downside protection through collars, or switch exposure based on regime evidence?

This report gives the first reproducible answer. It measures the market state, classifies the volatility and drawdown regime, compares synthetic overlays, calculates active risk versus passive Brazilian equity, and turns the result into a portfolio decision frame.

4. Market State

The market-state layer gives the first context for Brazilian risk. Local equity behavior is not enough. External Brazil exposure, global equity pressure, USD/BRL, and VIX help explain whether the current movement is isolated noise or part of a wider stress channel.

The baseline market layer tracks Brazilian equity exposure, external Brazil exposure, global equity risk, USD/BRL, and VIX. This gives a first cross-market view of local beta, external Brazil risk, global risk appetite, currency stress, and volatility pressure.

This is not yet a full production model. It is a clean research base. The value is transparency. A reviewer can inspect the assumptions, rerun the output, and challenge the decision logic.

5. Data Provenance and Evidence Classification

This section protects the credibility of the work. A finished decision report must tell the reviewer what is observed, what is model-derived, what is synthetic, and what is only a rule-based signal.

A robust report must separate observed data from modeled signals.

The market layer uses real public market data downloaded through Yahoo Finance with yfinance. The derivatives layer is synthetic. Covered call and collar premiums are estimated through the Black-Scholes engine until real B3 listed-option chains are integrated.

This distinction matters. A real price series can support risk measurement. A synthetic option premium can support research design. It cannot yet support live execution decisions.

Layer	Item	Source	Evidence Type	Status
market_data	brazil_equity	Yahoo Finance through yfinance	real_public_market_price_series	real market proxy
market_data	brazil_external	Yahoo Finance through yfinance	real_public_market_price_series	real market proxy
market_data	global_equity	Yahoo Finance through yfinance	real_public_market_price_series	real market proxy
market_data	fx_usdbrl	Yahoo Finance through yfinance	real_public_market_price_series	real market proxy

Layer	Item	Source	Evidence Type	Status
market_data	vix	Yahoo Finance through yfinance	real_public_market_price_series	real market proxy
derived_metric	realized_volatility	BRAVO Lab calculation	model_derived	derived from real market data
derived_metric	drawdown	BRAVO Lab calculation	model_derived	derived from real market data
regime_signal	baseline_regime_classifier	BRAVO Lab rule-based classifier	model_generated_signal	decision signal, not observed market data
synthetic_derivatives	covered_call_overlay	BRAVO Lab synthetic option engine	synthetic_research_assumption	not real B3 option-chain evidence
synthetic_derivatives	collar_overlay	BRAVO Lab synthetic option engine	synthetic_research_assumption	not real B3 option-chain evidence
strategy_logic	stress_aware_overlay	BRAVO Lab switching rule	model_generated_strategy_rule	research rule requiring validation

6. Regime Diagnosis

The regime diagnosis classifies the local Brazilian equity environment before the stress-transmission layer is added. It uses realized volatility, volatility percentile, and drawdown to identify whether the overlay discussion is happening in calm, fragile, stress, or extreme-stress conditions.

The current Brazilian equity signal sits in stress.

That matters because the same overlay can behave well in one regime and poorly in another. A covered call can create useful income in a range-bound market, but it can also sell the recovery too cheaply. A collar can protect the book during stress, but it can also become expensive insurance if drawdown risk fades.

The regime classifier uses realized volatility, volatility percentile, and drawdown. It is simple by design. The point is to create an auditable baseline before adding GARCH, MTV-GARCH, stress transmission indexes, wavelets, CCA, or machine learning.

Regime Snapshot

Latest classified regime: stress

Latest realized volatility: 16.15%

Latest drawdown: -15.20%

Regime Distribution

Regime	Observations	Share
stress	921	31.03%
extreme_stress	737	24.83%
calm	674	22.71%
fragile	334	11.25%
neutral	302	10.18%

7. Baseline Risk Metrics

The baseline risk table is the first risk map. It shows whether the assets feeding BRAVO carry normal volatility, deep drawdown risk, tail risk, or unstable return patterns.

The table below gives the first risk layer across the monitored assets. It is not a final allocation model. It is the risk map used to decide whether the overlay discussion is taking place in a calm, fragile, or stressed environment.

Asset	Ann. Return	Ann. Volatility	Sharpe	Sortino	Max Drawdown	VaR 95%
brazil_equity	9.92%	23.14%	0.525	0.686	-46.93%	-2.21%
fx_usdbrl	6.28%	16.45%	0.452	0.698	-26.80%	-1.59%
brazil_external	2.57%	33.91%	0.246	0.323	-66.54%	-3.29%
global_equity	13.30%	16.92%	0.823	0.985	-33.72%	-1.60%
vix	1.92%	133.45%	0.621	1.271	-85.66%	-10.52%

Asset	Ann. Return	CVaR 95%	Obs.
brazil_equity	9.92%	-3.28%	3239
fx_usdbrl	6.28%	-2.23%	3239
brazil_external	2.57%	-4.84%	3239
global_equity	13.30%	-2.59%	3239
vix	1.92%	-14.20%	3239

8. Synthetic Overlay Results

This section compares passive exposure, covered-call income, collar protection, and stress-aware switching. The option overlays are synthetic, so the numbers are research evidence, not live tradability claims.

The first overlay engine compares four exposures:

- passive Brazilian equity exposure
- synthetic covered call overlay
- synthetic protective collar overlay
- stress-aware overlay switching

The engine uses a 21-trading-day rebalance approximation, synthetic Black-Scholes option premiums, and a transaction-cost assumption of 5.0 basis points per option leg.

The stress-aware overlay maps regimes into actions: passive exposure in calm conditions, covered call income in fragile conditions, and collar protection in stress or extreme-stress conditions. It does not claim live tradability. It is a controlled research baseline before adding real B3 option chains, taxes, liquidity, and execution constraints.

Returns in this section are approximately monthly strategy-period returns, annualized using 12.0 periods per year.

Strategy	Ann. Return	Ann. Volatility	Sharpe	Max Drawdown	Best Period
passive_brazil_equity	9.93%	22.24%	0.543	-37.77%	21.22%
covered_call	7.92%	16.80%	0.549	-36.79%	10.57%
collar	5.54%	10.23%	0.579	-21.49%	4.11%
stress_aware_overlay	6.79%	13.62%	0.552	-23.01%	10.58%

Strategy	Worst Period	Obs.
passive_brazil_equity	-35.92%	151
covered_call	-35.16%	151
collar	-4.76%	151

Strategy	Worst Period	Obs.
stress_aware_overlay	-14.33%	151

9. Active Risk Diagnostics

This section asks whether each overlay earns the right to deviate from passive Brazilian equity. Return alone is insufficient. Tracking error, information ratio, hit rate, downside hit rate, and worst active period tell the reviewer whether the active decision is governable.

Absolute return is not enough. A portfolio desk also needs to know whether the overlay earns enough to justify its deviation from passive Brazilian equity.

Tracking error measures how far the overlay moves away from passive exposure. The information ratio tests whether that active deviation is rewarded. Hit rate shows how often the overlay beats passive. Downside hit rate is stricter. It asks whether the overlay helps when passive exposure is already losing money.

Strategy	Ann. Active Return	Tracking Error	Information Ratio	Hit Rate	Downside Hit Rate
covered_call	-2.87%	10.33%	-0.278	74.83%	100.00%
collar	-6.16%	14.82%	-0.416	69.54%	100.00%
stress_aware_overlay	-4.57%	13.62%	-0.335	44.37%	64.18%

Strategy	Worst Active Period	Obs.
covered_call	-16.46%	151
collar	-17.62%	151
stress_aware_overlay	-17.62%	151

9. Active Risk by Regime

This is one of the most important sections in the report. A strategy can look acceptable across the full sample and still fail inside a specific regime.

Full-sample tracking error is useful, but it is not enough for portfolio governance. A strategy can look acceptable across the whole sample while creating too much active risk in a specific market state.

This diagnostic shows where each overlay creates tracking error versus passive Brazilian equity exposure: calm markets, fragile markets, stress markets, or extreme-stress markets.

Regime	Strategy	Annualized Active Return	Tracking Error	Information Ratio	Hit Rate	Downside Hit Rate
calm	covered_call	-17.41%	10.71%	-1.625	56.67%	100.00%
calm	collar	-23.87%	10.63%	-2.246	50.00%	100.00%
calm	stress_aware_overlay	-10.70%	9.17%	-1.167	26.67%	57.14%
extreme_stress	covered_call	1.49%	13.47%	0.111	85.29%	100.00%
extreme_stress	collar	7.39%	25.25%	0.293	82.35%	100.00%
extreme_stress	stress_aware_overlay	3.31%	24.49%	0.135	64.71%	75.00%
fragile	covered_call	-1.33%	7.28%	-0.183	76.19%	100.00%
fragile	collar	-6.61%	7.71%	-0.858	76.19%	100.00%
fragile	stress_aware_overlay	-2.02%	5.09%	-0.397	38.10%	55.56%

Regime	Strategy	Annualized Active Return	Tracking Error	Information Ratio	Hit Rate	Downside Hit Rate
neutral	covered_call	-4.74%	8.09%	-0.587	66.67%	100.00%
neutral	collar	-7.34%	10.12%	-0.725	66.67%	100.00%
neutral	stress_aware_overlay	-8.34%	7.37%	-1.131	9.52%	11.11%
stress	covered_call	3.69%	9.01%	0.410	82.22%	100.00%
stress	collar	-3.84%	9.60%	-0.400	71.11%	100.00%
stress	stress_aware_overlay	-5.86%	9.14%	-0.641	60.00%	81.82%

Regime	Strategy	Avg. Active Period	Best Active Period	Worst Active Period	Obs.
calm	covered_call	-1.45%	1.70%	-7.93%	30
calm	collar	-1.99%	0.59%	-8.93%	30
calm	stress_aware_overlay	-0.89%	1.70%	-8.23%	30
extreme_stress	covered_call	0.12%	7.34%	-16.46%	34
extreme_stress	collar	0.62%	31.28%	-17.62%	34
extreme_stress	stress_aware_overlay	0.28%	31.28%	-17.62%	34
fragile	covered_call	-0.11%	1.84%	-6.06%	21
fragile	collar	-0.55%	2.20%	-6.99%	21
fragile	stress_aware_overlay	-0.17%	2.20%	-3.53%	21
neutral	covered_call	-0.40%	2.84%	-7.95%	21
neutral	collar	-0.61%	4.10%	-8.91%	21
neutral	stress_aware_overlay	-0.70%	1.03%	-8.91%	21
stress	covered_call	0.31%	2.52%	-13.39%	45
stress	collar	-0.32%	4.20%	-14.07%	45
stress	stress_aware_overlay	-0.49%	1.83%	-14.07%	45

Active Risk by Regime Interpretation

This is one of the most important sections in the report. A strategy can look acceptable across the full sample and still fail inside a specific regime.

Active-risk-by-regime read: collar creates the highest tracking error in the extreme_stress regime. covered_call shows the strongest information ratio in the stress regime. This separates a strategy that looks attractive on average from a strategy that is governable under specific market states.

10. Multi-Asset Stress Signals

The multi-asset stress layer turns BRAVO from a simple local backtest into a stress-transmission framework. It identifies whether pressure is coming from Brazil drawdown, FX, external Brazil exposure, global equity pressure, or VIX.

The first regime layer is built from local Brazilian equity behavior. That is useful, but incomplete. A portfolio stress system should also look outside the local price series.

This section adds external Brazil exposure, global equity pressure, USD/BRL pressure, and VIX pressure to create a broader multi-asset stress dashboard. It does not forecast markets. It identifies where pressure is currently coming from.

Date	Composite Stress Score	Stress Regime	Top Pressure 1	Value 1	Top Pressure 2	Value 2
2026-06-09	0.345	calm	brazil_drawdown_pressure	1.877	global_equity_pressure	0.191

Date	Composite Stress Score	Top Pressure 3	Value 3	Brazil Drawdown	Brazil 21D Vol	VIX Level
2026-06-09	0.345	brazil_vol_pressure	0.000	-15.20%	16.15%	18.160

Multi-Asset Stress Interpretation

Multi-asset stress read: the latest composite stress score is 0.34, classified as calm. The strongest current pressure inputs are brazil_drawdown_pressure, global_equity_pressure, and brazil_vol_pressure. This extends the framework beyond local price behavior and starts moving BRAVO Lab toward a broader Brazil stress transmission dashboard.

11. Brazil Stress Transmission Index

BSTI is the central stress signal. It compresses multiple Brazil-related pressure channels into a 0 to 100 index while keeping the dominant channel visible. Its purpose is portfolio governance, not market prophecy.

The multi-asset stress dashboard shows the individual pressure inputs. The Brazil Stress Transmission Index converts those inputs into a formal composite index from 0 to 100.

The purpose is portfolio governance. The index gives the research a single monitorable stress state while preserving the underlying channels that explain where pressure is coming from.

Latest BSTI State

Date	BSTI 0-100	Raw Score	Regime	Stress Breadth	Active Channels	Dominant Channel
2026-06-09	13.466	0.404	calm	0.167	1	brazil_drawdown_pressure

Date	BSTI 0-100	Dominant Value	Top Channel 1	Value 1	Top Channel 2	Value 2
2026-06-09	13.466	1.877	brazil_drawdown_pressure	1.877	global_equity_pressure	0.191

Date	BSTI 0-100	Top Channel 3	Value 3
2026-06-09	13.466	brazil_vol_pressure	0.000

Historical BSTI Regime Distribution

BSTI Regime	Observations	Share
calm	1968	60.76%
fragile	989	30.53%
stress	224	6.92%
extreme_stress	58	1.79%

BSTI Interpretation

BSTI read: the latest Brazil Stress Transmission Index is 13.5 out of 100, classified as calm. Stress breadth is 0.17, with 1 active pressure channels. The dominant pressure channel is brazil_drawdown_pressure. The top three channels are brazil_drawdown_pressure, global_equity_pressure, and brazil_vol_pressure. This turns the multi-asset stress dashboard into a formal Brazil stress-transmission index that can be monitored, reported, and later tested against overlay decisions.

12. BSTI Threshold Validation

A stress index is useful only if its thresholds connect to future risk outcomes. This validation layer tests whether BSTI levels relate to future drawdowns and overlay behavior under elevated stress.

A stress index is only useful if its thresholds connect to portfolio-relevant outcomes. This validation layer tests whether BSTI levels are associated with future benchmark losses, future drawdowns, and overlay behavior when stress is already elevated.

This does not claim forecasting power. It tests whether BSTI is informative enough to support portfolio-governance discussion.

BSTI Thresholds versus Future Outcomes

Horizon	BSTI Threshold	Obs.	Signal Obs.	Signal Freq.	Avg Return Signal On	Avg Return Signal Off
21	15.000	3229	1263	39.11%	0.73%	1.23%
21	33.000	3229	281	8.70%	-0.50%	1.18%
21	50.000	3229	58	1.80%	-1.17%	1.07%
63	15.000	3229	1263	39.11%	3.61%	2.84%
63	33.000	3229	281	8.70%	2.39%	3.21%
63	50.000	3229	58	1.80%	1.65%	3.17%

Horizon	BSTI Threshold	Avg Max DD Signal On	Avg Max DD Signal Off	Neg Return Precision	5% DD Precision	10% DD Precision
21	15.000	-5.98%	-4.74%	44.81%	48.14%	10.61%
21	33.000	-8.09%	-4.95%	45.55%	58.01%	22.06%
21	50.000	-9.96%	-5.14%	44.83%	62.07%	31.03%
63	15.000	-10.27%	-9.02%	39.67%	85.67%	35.00%
63	33.000	-12.62%	-9.21%	43.06%	90.04%	45.20%
63	50.000	-14.94%	-9.41%	46.55%	98.28%	56.90%

Horizon	BSTI Threshold	5% DD Recall
21	15.000	43.80%
21	33.000	11.74%
21	50.000	2.59%
63	15.000	38.60%
63	33.000	9.03%
63	50.000	2.03%

Overlay Behavior When BSTI Is Elevated

BSTI Threshold	Strategy	Obs.	Avg Active Return	Annualized Active Return	Tracking Error	Information Ratio
15.000	covered_call	58	0.43%	5.16%	6.73%	0.767
15.000	collar	58	1.07%	12.78%	16.98%	0.753
15.000	stress_aware_overlay	58	0.64%	7.72%	16.10%	0.480
33.000	covered_call	9	0.78%	9.31%	1.51%	6.148
33.000	collar	9	5.63%	67.52%	35.16%	1.921
33.000	stress_aware_overlay	9	3.81%	45.68%	35.75%	1.278
50.000	covered_call	6	0.62%	7.44%	0.94%	7.951
50.000	collar	6	1.19%	14.27%	5.02%	2.845
50.000	stress_aware_overlay	6	0.07%	0.86%	0.61%	1.414

BSTI Threshold	Strategy	Hit Rate	Best Active Period	Worst Active Period
15.000	covered_call	91.38%	2.12%	-8.89%
15.000	collar	87.93%	31.28%	-9.19%
15.000	stress_aware_overlay	48.28%	31.28%	-8.89%
33.000	covered_call	100.00%	1.78%	0.30%
33.000	collar	100.00%	31.28%	0.16%
33.000	stress_aware_overlay	44.44%	31.28%	0.00%
50.000	covered_call	100.00%	0.98%	0.30%
50.000	collar	100.00%	3.61%	0.16%
50.000	stress_aware_overlay	16.67%	0.43%	0.00%

BSTI Validation Interpretation

BSTI validation read: threshold 50.0 produced the highest 5 percent drawdown-event precision over the 63 trading-day horizon. For overlays, covered_call showed the strongest information ratio when BSTI was above 50.0. This does not prove forecast power, but it starts testing whether the index is useful as a portfolio-governance warning signal rather than just a descriptive dashboard.

13. BSTI Threshold Calibration

Calibration asks whether different threshold and weighting choices create a more useful governance signal. The goal is not to overfit the index. The goal is to test whether the signal becomes more decision-useful under alternative designs.

The first BSTI threshold validation tests a fixed index structure. This calibration layer asks a more demanding question: does the stress index behave better under alternative thresholds and channel weights?

The point is not to overfit the index. The point is to test whether the chosen structure is robust enough for portfolio-governance discussion.

Calibration Grid

Weight Scheme	Horizon	Threshold	Governance Score	Signal Freq.	5% DD Precision	5% DD Recall
local_risk_heavy	21	50.000	0.343	3.59%	70.69%	5.91%
local_risk_heavy	21	10.000	0.339	55.62%	46.83%	60.59%
balanced	21	10.000	0.335	56.83%	46.10%	60.95%
fx_vix_heavy	21	10.000	0.328	54.04%	46.02%	57.85%
global_risk_heavy	21	10.000	0.325	50.82%	46.50%	54.97%
external_brazil_heavy	21	10.000	0.324	54.72%	45.33%	57.71%
local_risk_heavy	21	15.000	0.320	42.43%	48.25%	47.62%
external_brazil_heavy	21	50.000	0.314	1.83%	66.10%	2.81%
balanced	21	15.000	0.311	39.11%	48.14%	43.80%
local_risk_heavy	21	40.000	0.308	7.53%	61.32%	10.73%
local_risk_heavy	21	33.000	0.303	11.92%	57.92%	16.07%
external_brazil_heavy	21	15.000	0.302	36.79%	47.64%	40.78%

Weight Scheme	Horizon	Avg Max DD Signal On	Avg Max DD Signal Off	Obs.
local_risk_heavy	21	-10.56%	-5.02%	3229
local_risk_heavy	21	-5.70%	-4.63%	3229
balanced	21	-5.63%	-4.68%	3229
fx_vix_heavy	21	-5.64%	-4.73%	3229
global_risk_heavy	21	-5.71%	-4.72%	3229
external_brazil_heavy	21	-5.64%	-4.72%	3229
local_risk_heavy	21	-5.92%	-4.71%	3229
external_brazil_heavy	21	-9.96%	-5.13%	3229
balanced	21	-5.98%	-4.74%	3229
local_risk_heavy	21	-8.49%	-4.96%	3229
local_risk_heavy	21	-7.71%	-4.89%	3229
external_brazil_heavy	21	-5.87%	-4.84%	3229

Best Calibration by Horizon

Horizon	Weight Scheme	Threshold	Governance Score	Signal Freq.	5% DD Precision
21	local_risk_heavy	50.000	0.343	3.59%	70.69%
63	balanced	10.000	0.503	56.83%	86.43%

Horizon	5% DD Recall	Selection Rule
21	5.91%	max_governance_score
63	56.58%	max_governance_score

Calibration Interpretation

BSTI calibration read: the strongest candidate is the balanced weighting scheme with threshold 10.0 over the 63 trading-day horizon. Its governance score is 0.503. This does not mean the index is optimized for prediction. It means the project now tests whether different stress-channel weights and alert thresholds produce more useful portfolio-governance signals.

14. BSTI Overlay Policy Selection

This is where diagnosis becomes action. The BSTI rule maps stress levels into passive exposure, covered-call income, or collar protection.

The BSTI policy layer converts the stress index into an explicit portfolio action. This is where the project moves from diagnosis to governance.

The rule is intentionally simple:

- low BSTI: remain in passive Brazil equity exposure
- medium BSTI: use covered calls to collect option premium
- high BSTI: use collars to prioritize downside control

This does not claim to forecast returns. It tests whether a transparent stress signal can discipline overlay selection across passive exposure, covered calls, collars, and the existing local-regime stress-aware overlay.

Policy Selection Summary

Selected Strategy	Obs.	Share	Avg BSTI	Avg Selected Return	Avg Passive Return
collar	9	5.96%	51.053	-3.20%	-8.83%
covered_call	49	32.45%	21.890	-0.72%	-1.09%
passive_brazil_equity	93	61.59%	7.323	3.06%	3.06%

Selected Strategy	Avg Active Return	Positive Active Rate
collar	5.63%	100.00%
covered_call	0.37%	89.80%
passive_brazil_equity	0.00%	0.00%

Policy Performance Comparison

Strategy	Ann. Return	Ann. Vol.	Max DD	Ann. Active	Tracking Error	Info. Ratio
passive_brazil_equity	12.09%	22.24%	-37.77%	0.00%	0.00%	NA
covered_call	9.22%	16.80%	-36.79%	-2.87%	10.33%	-0.278
collar	5.92%	10.23%	-21.49%	-6.16%	14.82%	-0.416
stress_aware_overlay	7.52%	13.62%	-23.01%	-4.57%	13.62%	-0.335
bsti_policy_overlay	17.54%	17.67%	-25.01%	5.45%	10.18%	0.535

Strategy	Ann. Return	Hit Rate	Obs.
passive_brazil_equity	12.09%	0.00%	151
covered_call	9.22%	74.83%	151
collar	5.92%	69.54%	151

Strategy	Ann. Return	Hit Rate	Obs.
stress_aware_overlay	7.52%	44.37%	151
bsti_policy_overlay	17.54%	35.10%	151

Policy Interpretation

BSTI policy read: the BSTI-driven policy most often selected passive_brazil_equity. Its annualized active return is 5.45% with tracking error 10.18% and information ratio 0.54. This turns BSTI from a dashboard into a governable overlay-selection rule that can be compared against passive exposure, covered calls, collars, and the local-regime stress-aware overlay.

15. BSTI Signal Persistence and State Transitions

Persistence separates a real warning process from a random spike. A temporary stress print should not trigger the same response as a durable warning state.

A stress signal is not useful only because it moves. It is useful when it persists long enough to support a decision process.

This section tests whether the Brazil Stress Transmission Index behaves like a governance signal rather than a random dashboard number. It classifies each BSTI observation into normal, warning, or stress states, then studies transition probabilities, state duration, warning-to-stress escalation, and dominant pressure-channel rotation.

BSTI State Transition Matrix

From State	To State	Transitions	Probability
normal	normal	1518	77.17%
normal	stress	62	3.15%
normal	warning	387	19.67%
stress	normal	27	9.57%
stress	stress	117	41.49%
stress	warning	138	48.94%
warning	normal	422	42.67%
warning	stress	103	10.41%
warning	warning	464	46.92%

BSTI State Duration Summary

State	Episodes	Avg Duration	Median Duration	Max Duration	Avg BSTI	Max BSTI
normal	450	4.373	2.000	46.000	8.385	14.998
warning	525	1.884	1.000	15.000	21.475	32.983
stress	165	1.709	1.000	24.000	41.687	99.514

Warning-to-Stress Escalation

Warning Events	Lookahead Obs.	Escalation Rate	Stay Warning/Stress Rate
989	3	24.17%	57.33%

Dominant Pressure-Channel Transitions

From Channel	To Channel	Transitions	Probability
brazil_drawdown_pressure	brazil_drawdown_pressure	548	63.80%
brazil_drawdown_pressure	external_brazil_pressure	85	9.90%
brazil_drawdown_pressure	global_equity_pressure	74	8.61%
brazil_drawdown_pressure	fx_pressure	53	6.17%
brazil_drawdown_pressure	brazil_vol_pressure	50	5.82%
brazil_drawdown_pressure	vix_pressure	49	5.70%
brazil_vol_pressure	brazil_vol_pressure	502	74.04%
brazil_vol_pressure	external_brazil_pressure	61	9.00%
brazil_vol_pressure	global_equity_pressure	37	5.46%
brazil_vol_pressure	brazil_drawdown_pressure	35	5.16%
brazil_vol_pressure	vix_pressure	25	3.69%
brazil_vol_pressure	fx_pressure	18	2.65%

Transition Interpretation

BSTI transition read: warning episodes persisted for an average of 1.88 observations, while stress episodes persisted for an average of 1.71 observations. The warning-to-stress escalation rate over the selected lookahead window was 24.17%. This helps distinguish a one-period stress flash from a governable warning state that may justify portfolio review, hedge discussion, or overlay-policy activation.

16. Drawdown and Recovery Diagnostics

This section tests the real trade-off of protection. An overlay may help during drawdowns and still damage the rebound.

Drawdown diagnostics ask whether the overlay helps at different levels of benchmark pain. Recovery diagnostics ask the uncomfortable second question: does the hedge damage the portfolio when the market starts rebounding?

This matters because a protective overlay can be useful during the fall and still become expensive during the recovery.

Drawdown-Depth Summary

Drawdown Bucket	Strategy	Avg. Strategy Return	Avg. Benchmark Return	Avg. Active Return	Hit Rate	Downside Protection Rate
deep_drawdown	covered_call	-0.39%	-0.97%	0.58%	89.09%	100.00%
deep_drawdown	collar	-0.42%	-0.97%	0.55%	78.18%	100.00%
deep_drawdown	stress_aware_overlay	-0.77%	-0.97%	0.20%	72.73%	90.91%
moderate_drawdown	covered_call	-0.24%	-0.42%	0.18%	79.31%	100.00%
moderate_drawdown	collar	-0.36%	-0.42%	0.06%	79.31%	100.00%
moderate_drawdown	stress_aware_overlay	-0.69%	-0.42%	-0.27%	31.03%	31.25%
near_peak	covered_call	2.68%	4.00%	-1.32%	56.00%	100.00%
near_peak	collar	2.13%	4.00%	-1.87%	52.00%	100.00%
near_peak	stress_aware_overlay	3.07%	4.00%	-0.93%	28.00%	70.00%
shallow_drawdown	covered_call	0.63%	1.06%	-0.43%	76.47%	100.00%
shallow_drawdown	collar	0.10%	1.06%	-0.95%	76.47%	100.00%
shallow_drawdown	stress_aware_overlay	0.21%	1.06%	-0.85%	23.53%	12.50%

Drawdown Bucket	Strategy	Best Active Period	Worst Active Period	Obs.
deep_drawdown	covered_call	7.34%	-16.46%	55
deep_drawdown	collar	31.28%	-17.62%	55
deep_drawdown	stress_aware_overlay	31.28%	-17.62%	55
moderate_drawdown	covered_call	1.51%	-4.30%	29
moderate_drawdown	collar	4.20%	-4.92%	29
moderate_drawdown	stress_aware_overlay	2.20%	-4.92%	29
near_peak	covered_call	1.84%	-8.89%	50
near_peak	collar	0.62%	-9.19%	50
near_peak	stress_aware_overlay	0.98%	-8.89%	50
shallow_drawdown	covered_call	1.70%	-7.95%	17
shallow_drawdown	collar	0.59%	-8.91%	17
shallow_drawdown	stress_aware_overlay	1.70%	-8.91%	17

Recovery-Window Summary

Strategy	Avg. Strategy Return	Avg. Benchmark Return	Avg. Active in Recovery	Hit Rate in Recovery	Missed Recovery Rate	Best Active Recovery
covered_call	4.12%	4.93%	-0.81%	62.79%	37.21%	7.34%
collar	2.88%	4.93%	-2.05%	48.84%	51.16%	1.10%
stress_aware_overlay	3.00%	4.93%	-1.93%	37.21%	46.51%	1.43%

Strategy	Avg. Strategy Return	Worst Active Recovery	Obs.
covered_call	4.12%	-16.46%	43
collar	2.88%	-17.62%	43
stress_aware_overlay	3.00%	-17.62%	43

Drawdown-Recovery Interpretation

Drawdown-recovery read: covered_call shows the strongest average active behavior during deep benchmark drawdowns. collar shows the largest active drag during recovery windows. This is the key hedge-governance trade-off: protection can help during the fall, but it must not destroy too much of the rebound.

17. Regime and Stress-Window Diagnostics

Full-sample metrics can hide the real question. A strategy that looks strong in normal conditions may fail when the benchmark is under pressure.

This section tests the overlay behavior by regime and during stress windows. The goal is to identify whether the strategy helps when protection, income discipline, and active-risk control matter most.

Regime-Level Performance

Regime	Strategy	Avg. Period Return	Median Period Return	Best Period	Worst Period
calm	passive_brazil_equity	3.81%	3.29%	12.50%	-4.07%
calm	covered_call	2.36%	3.45%	4.78%	-3.45%
calm	collar	1.82%	3.19%	3.61%	-3.76%
calm	stress_aware_overlay	2.92%	3.13%	10.58%	-4.07%
extreme_stress	passive_brazil_equity	-1.04%	-1.04%	21.22%	-35.92%
extreme_stress	covered_call	-0.92%	0.18%	10.57%	-35.16%
extreme_stress	collar	-0.43%	-0.63%	4.11%	-4.65%
extreme_stress	stress_aware_overlay	-0.77%	-0.63%	4.11%	-14.33%
fragile	passive_brazil_equity	0.94%	0.49%	10.54%	-6.64%
fragile	covered_call	0.83%	1.36%	4.48%	-5.13%
fragile	collar	0.39%	0.90%	3.55%	-4.51%
fragile	stress_aware_overlay	0.77%	0.49%	10.54%	-5.27%
neutral	passive_brazil_equity	1.11%	0.66%	12.47%	-8.34%
neutral	covered_call	0.72%	1.62%	5.33%	-7.38%
neutral	collar	0.50%	1.11%	3.69%	-4.61%
neutral	stress_aware_overlay	0.42%	0.75%	7.68%	-8.34%
stress	passive_brazil_equity	0.67%	0.19%	17.55%	-8.72%
stress	covered_call	0.98%	1.59%	7.35%	-7.58%
stress	collar	0.35%	0.72%	3.90%	-4.76%
stress	stress_aware_overlay	0.18%	0.72%	3.90%	-7.58%

Regime	Positive Hit Rate	Obs.
calm	76.67%	30
calm	80.00%	30
calm	76.67%	30
calm	80.00%	30
extreme_stress	38.24%	34
extreme_stress	52.94%	34
extreme_stress	47.06%	34
extreme_stress	47.06%	34
fragile	57.14%	21
fragile	61.90%	21

Regime	Positive Hit Rate	Obs.
fragile	57.14%	21
fragile	57.14%	21
neutral	57.14%	21
neutral	61.90%	21
neutral	57.14%	21
neutral	57.14%	21
stress	51.11%	45
stress	62.22%	45
stress	53.33%	45
stress	53.33%	45

Stress-Window Summary

Strategy	Avg. Stress Return	Median Stress Return	Worst Stress Return	Best Stress Return	Hit Rate vs Passive
passive_brazil_equity	-0.07%	-0.37%	-35.92%	21.22%	NA
covered_call	0.16%	0.91%	-35.16%	10.57%	83.54%
collar	0.02%	0.17%	-4.76%	4.11%	75.95%
stress_aware_overlay	-0.23%	0.17%	-14.33%	4.11%	62.03%

Strategy	Downside Protection Rate	Obs.
passive_brazil_equity	NA	79
covered_call	100.00%	79
collar	100.00%	79
stress_aware_overlay	78.57%	79

Stress-Window Interpretation

Stress-window read: collar showed the strongest worst-period protection during stress windows, while covered_call showed the strongest average stress-period return. The portfolio question is whether the protection benefit is large enough to justify the active risk and implementation complexity.

18. Strategy Help-Hurt Diagnostics

The help-hurt diagnostic explains the trade-off behind each overlay. A strategy can protect the downside and still hurt the portfolio if it gives away too much rebound. It can also improve income while failing to protect the book when the benchmark is already falling.

This section separates the periods where each overlay adds active value from the periods where it creates drag versus passive Brazilian equity.

Strategy	Avg. Active Return	Active When Passive Up	Active When Passive Down	Hit Rate	Missed Upside Rate	Downside Protection Rate
covered_call	-0.24%	-1.32%	1.08%	74.83%	45.78%	100.00%
collar	-0.51%	-2.23%	1.60%	69.54%	55.42%	100.00%
stress_aware_overlay	-0.38%	-1.61%	1.13%	44.37%	38.55%	64.18%

Strategy	Avg. Active Return	Best Active Period	Worst Active Period	Primary Help	Primary Hurt	Obs.
covered_call	-0.24%	7.34%	-16.46%	downside_protection	missed_upside	151
collar	-0.51%	31.28%	-17.62%	downside_protection	missed_upside	151
stress_aware_overlay	-0.38%	31.28%	-17.62%	downside_protection	missed_upside	151

Help-Hurt Interpretation

Help-hurt read: covered_call currently shows the strongest downside protection behavior versus passive exposure. collar shows the highest missed-upside risk when passive Brazilian equity is positive. This is the central overlay trade-off: the portfolio can reduce left-tail pain, but protection and income strategies can also give away part of the rebound.

19. Implementation Drag Diagnostics

Implementation drag separates gross signal quality from net result. A strategy that looks good before friction can fail after transaction costs, taxes, liquidity, and roll mechanics.

The implementation-drag diagnostic separates the gross overlay signal from the net result after transaction costs. This matters because an overlay can look useful before costs and still fail after realistic rebalancing drag.

The diagnostic does not yet decompose every option leg into premium income, payoff effect, and moneyness attribution. It is the intermediate institutional layer showing whether the overlay signal survives implementation.

Strategy	Gross Active	Implementation Drag	Net Active	Total Drag	Drag / Gross Signal	Cost Survival
covered_call	-0.19%	-0.05%	-0.24%	-7.55%	0.265	1.265
collar	-0.41%	-0.10%	-0.51%	-15.10%	0.242	1.242
stress_aware_overlay	-0.32%	-0.06%	-0.38%	-8.85%	0.182	1.182

Strategy	Gross Active	Active When Passive Up	Active When Passive Down	Best Net Active	Worst Net Active	Obs.
covered_call	-0.19%	-1.32%	1.08%	7.34%	-16.46%	151
collar	-0.41%	-2.23%	1.60%	31.28%	-17.62%	151
stress_aware_overlay	-0.32%	-1.61%	1.13%	31.28%	-17.62%	151

Implementation Interpretation

Implementation read: covered_call currently shows the strongest average net active return after transaction costs. covered_call is the most cost-sensitive overlay by drag-to-gross signal ratio. This matters because an overlay that looks useful before costs can become weak once turnover, option-leg execution, and rebalancing drag are included.

20. Option Overlay Attribution

Option attribution breaks the overlay into economic parts: premium income, protection cost, payoff effect, implementation drag, and net effect.

This attribution layer explains why the synthetic option overlay worked or failed. It separates the model-implied return into premium income, protection cost, payoff effect, implementation drag, and net overlay effect.

This is not real B3 option-chain attribution. It is model-implied attribution based on the synthetic Black-Scholes overlay engine used in this project. That disclosure is important because it keeps the research honest while still making the derivatives logic inspectable.

Strategy	Avg. Underlying Return	Avg. Call Premium Income	Avg. Put Protection Cost	Avg. Option Payoff Effect	Avg. Implementation Drag	Avg. Gross Overlay Effect
collar	1.05%	1.30%	-1.23%	-0.73%	-0.10%	-0.67%
covered_call	1.05%	1.30%	0.00%	-1.63%	-0.05%	-0.33%
stress_aware_overlay	1.05%	0.94%	-0.78%	-0.72%	-0.06%	-0.57%

Strategy	Avg. Underlying Return	Avg. Net Overlay Effect	Avg. Net Strategy Return	Positive Net Overlay Rate	Obs.
collar	1.05%	-0.77%	0.28%	24.03%	154
covered_call	1.05%	-0.38%	0.67%	72.08%	154
stress_aware_overlay	1.05%	-0.63%	0.42%	20.78%	154

Option Attribution Interpretation

Option-attribution read: covered_call currently shows the strongest average net overlay effect after model-implied premium, payoff, protection cost, and implementation drag. collar generates the largest average call-premium income. stress_aware_overlay has the strongest average payoff contribution. This is still synthetic attribution, not real B3 option-chain attribution, but it makes the overlay engine explain why the strategy works or fails.

21. Option Attribution by Regime and Drawdown Bucket

The previous attribution table explains the average option overlay effect. This section asks where that effect appears. Premium income, protection cost, payoff effect, and implementation drag do not matter equally in every market state.

This layer separates the synthetic option attribution by regime and by benchmark drawdown bucket. The goal is to identify whether the strategy is earning income in calm markets, protecting in stress, paying off in deep drawdowns, or creating drag near market peaks.

Option Attribution by Regime

Context	Strategy	Avg. Underlying Return	Avg. Call Premium Income	Avg. Put Protection Cost	Avg. Option Payoff Effect	Avg. Implementation Drag
calm	collar	1.07%	0.82%	-0.77%	-0.82%	-0.10%
calm	covered_call	1.07%	0.82%	0.00%	-1.31%	-0.05%
calm	stress_aware_overlay	1.07%	0.00%	0.00%	0.00%	0.00%
extreme_stress	collar	2.00%	1.99%	-1.90%	-1.39%	-0.10%
extreme_stress	covered_call	2.00%	1.99%	0.00%	-2.89%	-0.05%
extreme_stress	stress_aware_overlay	2.00%	1.99%	-1.90%	-1.39%	-0.10%
fragile	collar	-1.21%	0.96%	-0.91%	0.38%	-0.10%
fragile	covered_call	-1.21%	0.96%	0.00%	-0.86%	-0.05%
fragile	stress_aware_overlay	-1.21%	0.96%	0.00%	-0.86%	-0.05%
neutral	collar	-0.08%	1.20%	-1.14%	0.14%	-0.10%
neutral	covered_call	-0.08%	1.20%	0.00%	-0.86%	-0.05%
neutral	stress_aware_overlay	-0.08%	0.00%	0.00%	0.00%	0.00%
stress	collar	1.82%	1.30%	-1.23%	-1.07%	-0.10%
stress	covered_call	1.82%	1.30%	0.00%	-1.58%	-0.05%
stress	stress_aware_overlay	1.82%	1.30%	-1.23%	-1.07%	-0.10%

Context	Strategy	Avg. Net Overlay Effect	Positive Net Overlay Rate	Obs.
calm	collar	-0.88%	18.18%	33
calm	covered_call	-0.54%	69.70%	33
calm	stress_aware_overlay	0.00%	0.00%	33
extreme_stress	collar	-1.40%	22.86%	35
extreme_stress	covered_call	-0.95%	60.00%	35
extreme_stress	stress_aware_overlay	-1.40%	22.86%	35
fragile	collar	0.34%	33.33%	18
fragile	covered_call	0.05%	83.33%	18
fragile	stress_aware_overlay	0.05%	83.33%	18
neutral	collar	0.10%	33.33%	24
neutral	covered_call	0.29%	87.50%	24
neutral	stress_aware_overlay	0.00%	0.00%	24
stress	collar	-1.11%	20.45%	44
stress	covered_call	-0.33%	70.45%	44
stress	stress_aware_overlay	-1.11%	20.45%	44

Option Attribution by Drawdown Bucket

Context	Strategy	Avg. Underlying Return	Avg. Call Premium Income	Avg. Put Protection Cost	Avg. Option Payoff Effect	Avg. Implementation Drag
deep_drawdown	collar	2.39%	1.91%	-1.82%	-1.91%	-0.10%
deep_drawdown	covered_call	2.39%	1.91%	0.00%	-2.62%	-0.05%
deep_drawdown	stress_aware_overlay	2.39%	1.81%	-1.73%	-1.79%	-0.10%
moderate_drawdown	collar	-0.26%	1.00%	-0.95%	0.19%	-0.10%
moderate_drawdown	covered_call	-0.26%	1.00%	0.00%	-0.80%	-0.05%
moderate_drawdown	stress_aware_overlay	-0.26%	0.65%	-0.42%	-0.72%	-0.06%
near_peak	collar	0.71%	0.90%	-0.85%	-0.51%	-0.10%
near_peak	covered_call	0.71%	0.90%	0.00%	-1.11%	-0.05%
near_peak	stress_aware_overlay	0.71%	0.36%	-0.15%	-0.31%	-0.03%
shallow_drawdown	collar	0.05%	1.05%	-0.99%	0.70%	-0.10%
shallow_drawdown	covered_call	0.05%	1.05%	0.00%	-1.63%	-0.05%
shallow_drawdown	stress_aware_overlay	0.05%	0.60%	-0.49%	1.21%	-0.05%
unknown	collar	0.81%	1.14%	-1.08%	-0.22%	-0.10%
unknown	covered_call	0.81%	1.14%	0.00%	-0.69%	-0.05%
unknown	stress_aware_overlay	0.81%	0.00%	0.00%	0.00%	0.00%

Context	Strategy	Avg. Net Overlay Effect	Positive Net Overlay Rate	Obs.
deep_drawdown	collar	-1.93%	25.93%	54
deep_drawdown	covered_call	-0.76%	66.67%	54
deep_drawdown	stress_aware_overlay	-1.80%	24.07%	54
moderate_drawdown	collar	0.14%	32.14%	28
moderate_drawdown	covered_call	0.15%	82.14%	28
moderate_drawdown	stress_aware_overlay	-0.53%	21.43%	28
near_peak	collar	-0.56%	18.00%	50
near_peak	covered_call	-0.25%	74.00%	50
near_peak	stress_aware_overlay	-0.13%	20.00%	50
shallow_drawdown	collar	0.66%	23.53%	17
shallow_drawdown	covered_call	-0.63%	64.71%	17
shallow_drawdown	stress_aware_overlay	1.28%	17.65%	17
unknown	collar	-0.26%	20.00%	5
unknown	covered_call	0.40%	80.00%	5
unknown	stress_aware_overlay	0.00%	0.00%	5

Option Context Interpretation

Option-context read: covered_call shows the strongest net overlay effect during stress regimes. collar generates the strongest call-premium income in calm regimes. stress_aware_overlay shows the strongest payoff contribution during deep drawdown buckets. collar shows the weakest net overlay effect near benchmark peaks. This separates income, protection, payoff, and drag across the market states where portfolio committees actually make allocation decisions.

22. Overlay Decision Matrix

The decision matrix turns the evidence into practical governance language: when a strategy is useful, when it is dangerous, and when the evidence remains unvalidated.

Strategy	Best Use	Main Risk	Portfolio Reading
Passive Brazilian Equity	Clean trend, calm regime, strong rebound	Full downside exposure	Use when upside participation matters more than protection
Covered Call	Fragile regime, elevated volatility, income objective	Upside is capped	Use when premium harvesting matters more than full upside capture
Collar	Stress regime, drawdown pressure, capital preservation	Protection cost and capped upside	Use when left-tail control matters more than return maximization
Stress-Aware Overlay	Regime-dependent switching	Model risk and signal timing	Uses passive in calm regimes, covered calls in fragile regimes, and collars in stress regimes

23. Strategy Trade-Off

Best annualized return: passive_brazil_equity

Best drawdown profile: collar

Best Sharpe profile: collar

Best information ratio versus passive: covered_call

Weakest drawdown profile: passive_brazil_equity

The decision is not to chase the highest return. The decision is to match the overlay to the regime and check whether active risk is rewarded.

Passive equity keeps the cleanest upside but carries the full left tail. Covered calls convert part of upside into premium income, which can help in sideways or moderately volatile markets. Collars give the book a defined protection logic, but their cost and upside cap must be justified by the current risk state. Stress-aware switching adds discipline, but only if the regime signal is stable enough to avoid unnecessary turnover.

24. Results SWOT

How to cope with the signal before turning it into a portfolio action.

This SWOT does not evaluate BRAVO Lab as a business project. It evaluates the current result as a portfolio decision signal.

The purpose is simple: separate what the model already shows from what still needs proof.

Dimension	Current Reading	Portfolio Meaning
Strengths	The report connects market state, regime classification, risk metrics, synthetic overlay performance, and active risk diagnostics in one reproducible pipeline.	The decision maker can compare passive exposure, covered call income, collar protection, and stress-aware switching under the same data window instead of reading each strategy in isolation.

Dimension	Current Reading	Portfolio Meaning
Weaknesses	The overlay premiums are synthetic. Real B3 option-chain behavior, liquidity, spreads, taxes, and execution costs are not yet included.	The current ranking should not be treated as live tradable evidence. It is a research signal that needs market microstructure validation.
Opportunities	The framework creates a path toward regime-aware overlay allocation with active-risk discipline. It can evolve from static comparison into a decision engine that changes exposure as volatility and drawdown conditions shift.	The strongest future use is not picking one permanent strategy. The value is learning when to hold beta, harvest premium, buy protection, or accept active risk.
Threats	The model can overstate strategy quality if synthetic option prices are too clean, if transaction costs are too low, or if full-sample performance hides stress-period failure.	The main risk is false confidence. A premium-looking result becomes dangerous if it is not tested across stress windows, costs, tracking error, and real option data.

SWOT Interpretation

The main strength is integration. The report does not leave the portfolio reviewer with isolated metrics. It links regime, risk, overlay behavior, and active deviation from the benchmark.

The main weakness is tradability. Synthetic option pricing is useful for building the research engine, but it is not enough for real portfolio deployment.

The main opportunity is regime switching with active-risk control. A static covered call or collar strategy is too rigid for Brazilian markets. The more valuable question is when the book should shift from participation to income to protection without taking unrewarded tracking error.

The main threat is clean-model illusion. A strategy can look strong before costs, spreads, liquidity, and stress subperiods. The next version must attack that weakness directly.

25. Transmission Read

Brazilian equity does not trade in isolation. The book can be hit through local rates, fiscal repricing, FX pressure, global volatility, commodity shocks, and external risk appetite.

This first version does not model every channel. It builds the base layer that future ShockBridge modules can extend into a formal Brazil Stress Transmission Index.

The key insight is simple: volatility is not only a number. It is a carrier of stress. When volatility rises with drawdown, the book is not just moving. It is absorbing transmission.

26. What To Watch Next

Watch whether drawdown stabilizes while realized volatility falls. If that does not happen, protection remains more valuable than income. If volatility falls without price repair, the market may be hiding fragility under calmer surface data.

The next model version should not simply add complexity. It should improve the decision. The immediate test is whether transaction costs, tracking error, and stress subperiod performance confirm or weaken the current overlay ranking.

27. What Would Break This View

This baseline view should be challenged if one of the following happens:

- 1. The regime classifier changes but overlay results do not respond.
- 2. Synthetic option premiums diverge too far from real B3 listed-option behavior.
- 3. Transaction costs erase the apparent benefit of the overlay.
- 4. The collar improves drawdown but destroys too much participation.

- 5. The covered call improves income but systematically sells the strongest rebounds.
- 6. The model performs well in the full sample but fails in stress subperiods.
- 7. The information ratio is positive in the full sample but weak during stress windows.
- 8. Tracking error rises without clear drawdown reduction or active return compensation.

28. Model Limits and Governance

The limits section prevents overclaiming. BRAVO is a decision-support research prototype. The synthetic option layer needs real B3 option-chain validation before production use.

This report is intentionally clear about what it does not prove.

- Yahoo Finance data is used as the baseline public-data source.
- No real B3 option-chain data is used yet.
- Option premiums are synthetic Black-Scholes approximations.
- Transaction costs, taxes, spreads, and liquidity constraints are not included yet.
- GARCH and MTV-GARCH are not yet active in this report.
- The regime classifier is transparent but not final.
- Active risk diagnostics are useful but still require stress-window validation.
- This is research infrastructure, not investment advice.

MODEL LIMITS

Model limits and final decision read

BRAVO is a research prototype for decision support. It uses public market data, model-derived diagnostics, and synthetic option-overlay assumptions. The project is not investment advice and it is not production trading infrastructure yet.

The next validation step is real B3 option-chain integration: observed strikes, maturities, bid-ask spreads, liquidity filters, transaction costs, taxes, roll rules, and mandate constraints.

The technical reproducibility materials, code, processed outputs, and figure-generation logic are available in the project repository. They are intentionally not listed inside this finished executive report.

The value of the project is the decision chain: transforming Brazilian market data into stress diagnostics, comparing overlay behavior, evaluating active risk, validating a stress index, defining policy selection, and communicating the result as a governance memo.